

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K^* \rho)_\tau f) \\ f(t=0) &= f_0 \end{cases}$$

(dV, Bruna & Burger, SIAM J. Appl. Dyn. Syst., 2025):

- PDE well-posed up to any time for singular Newtonian ∇K and $\sigma_x \geq 0$ and any τ

- No blow-up in infinite time in L^∞ using (Alikakos, CPDE, 1979)

Analysis

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(dW, Bruna & Burger, SIAM J. Appl. Dyn. Syst., 2025):

- PDE well-posed up to any time for singular Newtonian ∇K and $\sigma_x > 0$ and any τ
- No blow-up in infinite time in L^∞ using (Alikakos, CPDE, 1979)

Analysis

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- What happens as $t \rightarrow \infty$?